

HemaLens

Offline AI-powered red blood cell morphology screening, running entirely in the browser.

The Problem

Diagnosing conditions like Sickle Cell Disease, Thalassemia, and various anemias depends on recognizing red blood cell morphology under a microscope — a skill that takes years to train. In remote and low-resource regions, a severe shortage of trained hematologists means this diagnostic step is often simply unavailable at the point of care.

Our Solution

HemaLens classifies red blood cells into 13 morphological classes directly from a smear image, entirely client-side — no server, no internet connection required after the page loads, and no patient image data ever leaves the device. A two-stage pipeline (MONAI U-Net segmentation, then EfficientNet-B0 classification) runs in-browser via ONNX Runtime Web.

How It Works

1. Smear image → **2. U-Net segmentation** locates individual cells → **3. EfficientNet-B0** classifies each cell crop → **4. Annotated result** with bounding boxes, class labels, and confidence scores. Both models run as ONNX graphs inside the browser's WebAssembly runtime — there is no backend inference server in this pipeline at all.

Key Results

Dataset	Chula-RBC-12 — 623 labeled smear images, 22,106 labeled cells, ~34:1 class imbalance
Classifier	EfficientNet-B0 + Focal Loss — 0.857 macro F1 on held-out validation data
Segmentation	MONAI U-Net, weakly-supervised — IoU ~0.26 (documented limitation, see repo)
Deployment	Fully offline, in-browser inference via ONNX Runtime Web — zero server cost

Why It Matters

By running fully offline with zero recurring server cost, HemaLens extends a capability normally gated behind specialist hematologists and lab infrastructure to any rural clinic with a smartphone or basic laptop — directly addressing the gap between diagnostic need and diagnostic access, with no patient image data ever transmitted off the device.

Tech Stack & What's Next

Stack: PyTorch · MONAI · EfficientNet-B0 · ONNX / ONNX Runtime Web · React. **Next:** stronger segmentation supervision (real annotated boundaries, not pseudo-masks), on-device performance tuning for lower-end hardware, and a guided capture flow for smartphone-to-microscope imaging.

Team: Olalere Robiu Gbolahan (ML/model architecture) & [Teammate name] (web/deployment) — University of Ilorin, Nigeria. | UnivaBio Hackathon (BioCatalysis × UnivaDev), Devpost.

This is a research/educational morphology classifier, not a certified diagnostic device. Clinical interpretation requires a qualified professional.